

- **Duowell (Telmisartan 40MG/Rosuvastatin 10MG) Film Coated Tablet**
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-

[Composition] Each tablet contains

Duowell Tablet 40/10mg: Telmisartan 40mg, Rosuvastatin calcium 10.4mg(10mg as Rosuvastatin)

Duowell Tablet 40/20mg: Telmisartan 40mg, Rosuvastatin calcium 20.8mg(20mg as Rosuvastatin)

Duowell Tablet 80/10mg: Telmisartan 80mg, Rosuvastatin calcium 10.4mg(10mg as Rosuvastatin)

Duowell Tablet 80/20mg: Telmisartan 80mg, Rosuvastatin calcium 20.8mg(20mg as Rosuvastatin)

[Description]

Duowell Tablet 40/10mg: Light orange, oval, film-coated tablet.

Duowell Tablet 40/20mg: Light pink, oval, film-coated tablet.

Duowell Tablet 80/10mg: Light orange, oval, film-coated tablet.

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[Therapeutic Indications]

Duowell Tablet should be used in patients treated simultaneously with both ingredients (Telmisartan and Rosuvastatin).

Telmisartan

1. Essential Hypertension

2. Risk reduction of Cardiovascular disease

Telmisartan is indicated to reduce the risk of death from myocardial infarction, stroke and cardiovascular disease in patients 55 years or older at high risk of developing major cardiovascular events who cannot tolerate an angiotensin converting enzyme inhibitor (ACEI).

High risk of cardiovascular events includes evidence of coronary artery disease, peripheral arterial disease, stroke, transient ischemic attack, or diabetes mellitus with evidence of end-organ damage.

Rosuvastatin

1. Primary Hypercholesterolemia (Type IIa including heterozygous familial hypercholesterolemia) and combined hyperlipidemia (Type IIb): Adjuvant to diet when uncontrollable by diet and exercise

2. Adjuvant to diet for homozygous familial hypercholesterolemia or other lipid-lowering treatments (e.g., LDL apheresis)

3. Slowing progression of atherosclerosis through lowering total-C and LDL-C to target level in hypercholesterolemia patients

4. Adjuvant to diet of the patients who have primary dysbetalipoproteinemia (Type III)
5. Risk reduction of cardiovascular disease: in individuals without clinically evident coronary heart disease but age ≥ 50 years old in men and ≥ 60 years old in women with hsCRP (high sensitive C-reactive protein) ≥ 2 mg/L and with at least one additional cardiovascular disease risk factor (ex: hypertension, low HDL-cholesterol, smoking or a family history of premature coronary heart disease etc.)
 - reduce the risk of stroke.
 - reduce the risk of myocardial infarction.
 - reduce the risk of arterial revascularization.

[Dosage and Administration]

This drug is administered only to adults, and the dose should be individualized to each patient based on the effectiveness and tolerability of telmisartan/rosuvastatin.

This drug can be taken once daily in the regular time (e.g. morning) if possible.

The fixed- dose combination is not suitable for initial therapy.

If you are taking both telmisartan and rosuvastatin, you can switch to this combination product having the same content of each active ingredient for convenience.

If a dosage adjustment is necessary, titration should be done with the individual components.

Patients with essential hypertension

Adults: The usually effective dose is telmisartan 40 mg once daily. Some patients may already benefit at a daily dose of telmisartan 20 mg. In cases where the target blood pressure is not achieved, the dose of telmisartan can be increased to a maximum of 80 mg once daily. Alternatively, telmisartan may be used in combination with thiazidetype diuretics such as hydrochlorothiazide which has been shown to have an additive blood pressure lowering effect with telmisartan. When considering raising the dose of telmisartan, it must be borne in mind that the maximum antihypertensive effect is generally attained four to eight weeks after the start of treatment.

Patients with cardiovascular disease

Adults: The recommended dose is telmisartan 80 mg once daily. It is not known whether doses lower than 80 mg of telmisartan are effective in reducing cardiovascular morbidity. When initiating telmisartan therapy for the reduction of cardiovascular morbidity, close monitoring of blood pressure is recommended, and if appropriate adjustment of medications that lower blood pressure may be necessary.

Patients with hepatic impairment

This drug should be used with caution in patients with moderate hepatic impairment, and the posology should not exceed telmisartan 40 mg once daily.

Patients with hyperlipidemia and hypercholesterolemia

Applies to patients with primary hypercholesterolemia (Type IIa including heterozygous familial hypercholesterolemia) and combined hyperlipidemia (Type IIb), primary dysbetalipoproteinemia (Type III) and homozygous familial hypercholesterolemia.

Prior to and during administration of rosuvastatin, patients should have a standard low-cholesterol diet and it must be maintained during the period of treatment. The initial dose is rosuvastatin 5 mg once a day, and if further reduction of LDL-cholesterol level is required, it can be adjusted to the maintenance dose and administered. The maintenance dose is rosuvastatin 10 mg once a day, and almost all patients are controlled in this dose. The maintenance dose should be adjusted properly according to LDL-cholesterol level, treatment target and patient's response with an interval of 4 weeks or longer, and can be increased to the maximum of rosuvastatin 20 mg a day.

Patients with renal insufficiency

No dose adjustment is necessary in patients with mild to moderate renal impairment. Rosuvastatin is contraindicated in patients with severe renal impairment. The use of rosuvastatin 20mg for the patients with moderate renal impairment should be carefully considered.

Patients with hepatic impairment

There was no increase in systemic exposure to rosuvastatin in subjects with Child-Pugh scores of 7 or below. However, increased systemic exposure has been observed in subjects with Child-Pugh scores of 8 and 9. In these patients an assessment of renal function should be considered. There is no experience in subjects with Child-Pugh scores above 9. Rosuvastatin is contraindicated in patients with active liver disease.

Patients with pre-disposing factors to myopathy

The recommended start dose is rosuvastatin 5 mg in patients with predisposing factors to myopathy. The rosuvastatin 40 mg dose is contraindicated in patients with pre-disposing factors for myopathy.

Use in patients of Asian ancestry

Increased systemic exposure of rosuvastatin has been seen in Asian subjects. The recommended start dose is rosuvastatin 5 mg for patients of Asian ancestry. The dose of rosuvastatin 40 mg is contraindicated.

Use in the elderly

Excessive hypotension is generally not desirable for the elderly (cerebral infarction may occur). When administered to the elderly, this medicine should be administered with caution while observing the patient's condition.

Others

As a result of the pharmacokinetic drug-drug interaction study between telmisartan 80mg and rosuvastatin 20mg, C_{max} and AUC of telmisartan in the effect of rosuvastatin increased 1.35-fold and 1.17-fold, respectively, and C_{max} and AUC of rosuvastatin in the effect of telmisartan increased 2-fold and 1.18-fold, respectively. Therefore, when this drug is administered, the potential risk caused by increased C_{max} of rosuvastatin should be evaluated, and patient should be carefully monitored.

[Route of Administration]

For oral administration only.

[Pharmacokinetics]

A biological equivalence study was conducted to compare the bioavailability of this drug (telmisartan/rosuvastatin 80/20 mg) and co-administration of telmisartan 80 mg and rosuvastatin 20 mg. As a result of measuring the concentrations of telmisartan and rosuvastatin in the blood through a single-dose, 2x2 cross-over study to healthy adults, the 90% confidence interval of the geometric mean ratio of the study drug to the comparator was within 0.80-1.25 for both AUC_{last} and C_{max}, satisfying the equivalence criterion.

Other pharmacokinetic profiles such as plasma-protein binding, distribution, biotransformation, and excretion of the Duowell tablet have not been studied separately, but the pharmacokinetic properties of each previously approved ingredient are as follows.

Telmisartan

1. Absorption

Following oral administration, peak concentrations of telmisartan are reached in 0.5 to 1 hour after dosing. Food slightly reduces the bioavailability of telmisartan, with a reduction in the area under the plasma concentration-time curve (AUC) of about 6% with the 40 mg and about 20% after a 160 mg dose. The absolute bioavailability of telmisartan is dose dependent. At 40 mg and 160 mg, the bioavailability was 42% and 58%, respectively. The pharmacokinetics of orally administered telmisartan are nonlinear over the dose range 20 mg to 160 mg, with greater than proportional increases of plasma concentrations (C_{max} and AUC) with increasing doses. Telmisartan shows bi-exponential decay kinetics with a terminal elimination half-life of approximately 24 hours. Trough plasma concentrations of telmisartan with once daily dosing are about 10% to 25% of peak plasma concentrations. Telmisartan has an accumulation index in plasma of 1.5 to 2.0 upon repeated once daily dosing.

2. Distribution

Telmisartan is highly bound to plasma proteins (>99.5%), mainly albumin and α₁ - acid glycoprotein. Plasma protein binding is constant over the concentration range achieved with recommended doses. The volume of distribution for telmisartan is approximately 500 liters indicating additional tissue binding.

3. Metabolism and Elimination

Following either intravenous or oral administration of ¹⁴C-labeled telmisartan, most of the administered dose (>97%) was eliminated unchanged in feces via biliary excretion; only minute amounts were found in the urine (0.91% and 0.49% of total radioactivity, respectively). Telmisartan is metabolized by conjugation to form a pharmacologically inactive acyl glucuronide; the glucuronide of the parent compound is the only metabolite that has been identified in human plasma and urine. After a single dose, the glucuronide represents approximately 11% of the measured radioactivity in plasma. The cytochrome P450 isoenzymes are not involved in the metabolism of telmisartan. Total plasma clearance of telmisartan is >800 mL/min. Terminal half-life and total clearance appear to be independent of dose.

Rosuvastatin

1. Absorption

In clinical pharmacology studies in man, peak plasma concentrations of rosuvastatin were reached 3 to 5 hours following oral dosing. Both C_{max} and AUC increased in approximate proportion to the dose. The absolute bioavailability of rosuvastatin is approximately 20%. The AUC of rosuvastatin does not differ following evening or morning drug administration. Administration of rosuvastatin with food did not affect the AUC of rosuvastatin.

2. Distribution

Mean volume of distribution at steady-state of rosuvastatin is approximately 134 liters. Rosuvastatin is 88% bound to plasma proteins, mostly albumin. This binding is reversible and independent of plasma concentrations.

3. Metabolism

Rosuvastatin is not extensively metabolized; approximately 10% of a radiolabeled dose is recovered as metabolite. The major metabolite is N-desmethyl rosuvastatin, which is formed principally by cytochrome P450/2C9, and in vitro studies have demonstrated that N-desmethyl rosuvastatin has approximately one-sixth to one-half the HMG-CoA reductase inhibitory activity of the parent compound. Overall, greater than 90% of active plasma HMG-CoA reductase inhibitory activity is accounted for by the parent compound.

4. Excretion

Following oral administration, rosuvastatin and its metabolites are primarily excreted in the feces (90%). After an intravenous dose, approximately 28% of total body clearance was via the renal route, and 72% by the hepatic route. The elimination half-life of rosuvastatin is approximately 19 hours.

[Pharmacodynamics]

ATC code: C10BX20

Mechanism of action

Telmisartan

Telmisartan is an orally active and specific angiotensin II receptor (type AT1) antagonist. Telmisartan displaces angiotensin II with very high affinity from its binding site at the AT1 receptor subtype, which is responsible for the known actions of angiotensin II. Telmisartan does not exhibit any partial agonist activity at the AT1 receptor. Telmisartan selectively binds the AT1 receptor. The binding is long-lasting.

Rosuvastatin

Rosuvastatin is a selective and competitive inhibitor of HMG-CoA reductase, the rate-limiting enzyme that converts 3-hydroxy-3-methylglutaryl coenzyme A to mevalonate, a precursor for cholesterol. The primary site of action of rosuvastatin is the liver, the target organ for cholesterol lowering.

Rosuvastatin increases the number of hepatic LDL receptors on the cell-surface, enhancing uptake and catabolism of LDL and it inhibits the hepatic synthesis of VLDL, thereby reducing the total number of VLDL and LDL particles.

The safety and efficacy of this drug (telmisartan/rosuvastatin 80/20 mg) was evaluated in a double-blind, randomized, multi-center phase III clinical study in patients with hypertension and hyperlipidemia. Therapeutic lifestyle change (TLC) was performed for 4~12 weeks during the screening period in patients prior to administration of the drug. If applicable, patients were asked to discontinue anti-hyperlipidemic drug completely during the screening period, and anti-hypertensive for at least two weeks before administration of the drug. After the therapeutic lifestyle change period, a total of 210 patients who satisfied the inclusion/exclusion criteria were randomized to the test group (telmisartan/rosuvastatin 80/20 mg: 84 patients), control group 1 (Rosuvastatin 20 mg: 42 patients), control group 2 (Telmisartan 80 mg: 43 patients) and control group 3 (Placebo: 41 patients). The drug was administered once a day for 8 weeks. Among them, the safety was evaluated for 208 patients who were randomized and administered at least 1 dose of the drug, and efficacy was evaluated for 203 patients who measured primary efficacy endpoints more than once. As co-primary efficacy parameters, the changes from baseline in MSDBP and LDL-C after 8 weeks of administration of the drug were evaluated. The telmisartan group (test group and control group 2) showed a statistically superior MSDBP lowering effect compared to non-telmisartan group (control group 1 and 3), and the test group showed a statistically superior MSDBP lowering effect compared to control group 1. The rosuvastatin group (test group and control group 1) showed a statistically superior LDL-C lowering effect compared to non-rosuvastatin group (control group 2 and 3), and the test group showed a statistically superior LDL-C lowering effect compared to control group 2.

Change from baseline in MSDBP (mmHg) after 8 weeks of administration of the drug

Change from baseline in MSDBP	Total		Among the group containing rosuvastatin	
	Telmisartan Group (N=123) ¹	Non-Telmisartan Group (N=80) ²	Telmisartan Group (N=80) ³	Non-Telmisartan Group (N=40) ⁴
LS Mean(SE)	-7.97(0.80)	-1.17(1.00)	-8.79(1.00)	-1.60(1.41)
Difference between LS Means(SE) [95% CI]	-6.81 [-9.34, -4.27]		-7.19 [-10.61, -3.77]	

¹Test group (Telmisartan/Rosuvastatin 80/20 mg) and Control group 2 (Telmisartan 80 mg)

²Control group 1 (Rosuvastatin 20 mg) and Control group 3 (Placebo)

³Test group (Telmisartan/Rosuvastatin 80/20 mg)

⁴Control group 1 (Rosuvastatin 20 mg)

MSDBP = mean seating diastolic blood pressure

LS Mean = least square mean

SE = standard error

Percent change from baseline in LDL-C (mg/dL) after 8 weeks of administration of the drug

Percent change from baseline in LDL-C	Total		Among the group containing Telmisartan	
	Rosuvastatin Group (N=119) ¹	Non- Rosuvastatin Group (N=82) ²	Rosuvastatin Group (N=80) ³	Non- Rosuvastatin Group (N=43) ⁴
LS Mean(SE)	-49.49(1.79)	0.58(2.15)	-49.34(2.19)	1.45(2.99)
Difference between LS Means(SE) [95% CI]	-50.06 [-55.58, -44.54]		-50.79 [-58.09, -43.48]	

¹Test group (Telmisartan/Rosuvastatin 80/20 mg) and Control group 1 (Rosuvastatin 20 mg)

²Control group 2 (Telmisartan 80 mg) and Control group 3 (Placebo)

³Test group (Telmisartan/Rosuvastatin 80/20 mg)

⁴Control group 2 (Telmisartan 80 mg)

LDL-C = low density lipoprotein cholesterol

LS Mean = least square mean

SE = standard error

[Special warnings and precautions for use]

1. Warnings

Drug that act directly on the renin-angiotensin-aldosterone system can cause fetal and neonatal morbidity and death when administered to pregnant women. Several dozen cases have been reported in the world literature in patients who were taking ACE inhibitors. When pregnancy is detected, Duowell Tablet should be discontinued as soon as possible. (Please refer to '7. Fertility, pregnancy and lactation')

2. It should be administered carefully to the following patients.

- 1) The elderly
 - 2) Patients with hyperkalemia or pre-disposing factors for high levels of potassium in the blood.
 - 3) Patients with moderate and mild hepatic impairment. (Monitor carefully, and the posology should not exceed 40 mg once daily.)
 - 4) As with other HMG-CoA reductase inhibitors, this medicine should be used with caution in patients who consume excessive quantities of alcohol and/or have a history of liver disease. It is recommended that liver function tests be carried out prior to the initiation of treatment. In patients with secondary hypercholesterolaemia caused by hypothyroidism or nephrotic syndrome, the underlying disease should be treated prior to initiating therapy with this medicine.
 - 5) Patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy.
 - 6) Patients with ischaemic cardiopathy, ischaemic cardiovascular disease or cerebrovascular accident. (Excessive reduction of blood pressure could result in a myocardial infarction or stroke.)
 - 7) Patients with gastroduodenal diseases such as active gastric or duodenal ulcer.
 - 8) Patients with severe hypotension and renal insufficiency (patients with bilateral renal artery stenosis or stenosis of the artery to a single functioning kidney)
 - 9) Patients with severe renal insufficiency (serum creatinine ≥ 3.0 mm/dL)
 - 10) Dual blockade of the renin-angiotensin-aldosterone system (RAAS): The combined use of ACE-inhibitors and angiotensin II receptor blockers or aliskiren is not recommended.
 - 11) Effects on skeletal muscle as with other HMG-CoA reductase inhibitors. e.g. myalgia, myopathy and, rarely, rhabdomyolysis have been reported in rosuvastatin-treated patients. There have been very rare reports of an immune-mediated necrotising myopathy (IMNM) during or after treatment with statins, including rosuvastatin. IMNM is clinically characterized by proximal muscle weakness and elevated serum creatine kinase (CK), which persist despite discontinuation of statin treatment.
 - As with other HMG-CoA reductase inhibitors, should be prescribed with caution in patients with pre-disposing factors for myopathy/rhabdomyolysis. Such factors include:
 - Renal impairment
 - Hypothyroidism
 - Personal or family history of hereditary muscular disorders
 - Previous history of muscular toxicity with another HMG-CoA reductase inhibitor or fibrate
 - Alcohol abuse
 - Age >70 years
 - Situations where an increase in plasma levels may occur
 - Concomitant use of fibrates.
- In such patients the risk of treatment should be considered in relation to possible benefit and clinical monitoring is recommended. If CK levels are significantly elevated at baseline (>5xULN) treatment should not be started.
- Patients should be asked to report inexplicable muscle pain, weakness or cramps immediately, particularly if associated with malaise or fever. CK levels should be measured in these patients.

Therapy should be discontinued if CK levels are markedly elevated ($>5\times\text{ULN}$) or if muscular symptoms are severe and cause daily discomfort (even if CK levels are $\leq 5\times\text{ULN}$). If symptoms resolve and CK levels return to normal, then consideration should be given to re-introducing rosuvastatin or an alternative HMG-CoA reductase inhibitor at the lowest dose with close monitoring. Routine monitoring of CK levels in asymptomatic patients is not warranted.

- Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of a plausible alternative cause of CK increase which may confound interpretation of the result. If CK levels are significantly elevated at baseline ($>5\times\text{ULN}$) a confirmatory test should be carried out within 5~7 days. If the repeat test confirms a baseline CK $>5\times\text{ULN}$, treatment should not be started.

12) This medicine should not be used in any patient with an acute, serious condition suggestive of myopathy or predisposing to the development of renal failure secondary to rhabdomyolysis. (e.g. sepsis, hypotension, major surgery, trauma, severe metabolic, endocrine and electrolyte disorders; or uncontrolled seizures)

13) In clinical trials there was no evidence of increased skeletal muscle effects in the small number of patients dosed with rosuvastatin and concomitant therapy. However, an increase in the incidence of myositis and myopathy has been seen in patients receiving other HMG-CoA reductase inhibitors together with fibric acid derivatives including gemfibrozil, ciclosporin, nicotinic acid, azole antifungals, protease inhibitors and macrolide antibiotics. Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors. Therefore, the combination of this medicine and gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of rosuvastatin with fibrates or niacin should be carefully weighed against the potential risks of such combinations. The 40 mg dose is contraindicated with concomitant use of a fibrate.

14) Proteinuria, detected by dipstick testing and mostly tubular in origin, has been observed in patients treated with higher doses of rosuvastatin, in particular 40 mg, where it was transient or intermittent in most cases. Proteinuria has not been shown to be predictive of acute or progressive renal disease. An assessment of renal function should be considered during routine follow-up of patients treated with a dose of 40 mg.

15) Pharmacokinetic studies show an approximate 2-fold elevation in median AUC and C_{max} in Asian subjects compared with Caucasians. Rosuvastatin dosage should be adjusted in Asian patients.

3. General precautions

As a result of drug interaction studies between 80 mg of telmisartan and 20 mg of rosuvastatin, the C_{max} and AUC of telmisartan in the presence of rosuvastatin increased about 1.35 times and 1.17 times, respectively, and the C_{max} of rosuvastatin in the presence of telmisartan increased about twice and the AUC of that increased about 1.18 times. Therefore, the potential risk of C_{max} increase in rosuvastatin should be assessed and the patient should be monitored closely.

Rosuvastatin single-agent clinical trials shows that musculoskeletal toxicity (including myalgia, myopathy (including myositis), and rarely rhabdomyolysis, etc.) with or without acute renal failure

in all dose-treated patients (especially greater than 20 milligrams). And CK levels were dose-dependently increased in patients receiving rosuvastatin. Use with caution in patients who are susceptible to rhabdomyolysis. CK levels should be measured before administration of this medicine. The baseline value can be used as a reference to determine whether the CK level has increased during the administration of this medicine. CK is not measured after intense exercise or other potential factors that may cause CK elevation because of difficulty in CK numerical analysis. If the CK value is significantly increased by 5 times the upper limit of normal at baseline, it should be measured again after 5 to 7 days to confirm the results. Treatment should not be started with this medicine if the CK level has increased significantly 5 times the upper limit of normal at baseline, even after 5 to 7 days. Periodic monitoring of CK levels is recommended after administration of this medicine. Also, patients who start treatment with rosuvastatin or who increase rosuvastatin dose should be informed of the risk of myopathy. If there is muscle pain, tenderness, muscular weakness or muscle spasms that cannot be accurately explained during the administration of rosuvastatin, you should advise your patients to consult your doctor immediately. In this case, the CK level should be measured and the drug should be discontinued if the CK level is significantly increased (> 5 times the normal upper limit). And discontinuation may also be determined by considering whether the severity of muscular symptoms and daily discomfort are caused regardless of the CK level. The lowest doses of rosuvastatin, as with other HMG-CoA reductase inhibitors, should be prescribed with caution in patients with pre-disposing factors for myopathy/rhabdomyolysis. Re-administration of this drug is not recommended if a suspected genetic myopathy is suspected. Also, the levels of transaminase (AST, ALT) should be monitored prior to administration of this medicine and monitored periodically after administration of this medicine. Rosuvastatin should be prescribed with caution in patients with elevated levels of transaminase and should be discontinued if the value of AST, SGOT, ALT or SGPT increases above 3 times the upper limit of normal.

Telmisartan

1) Renal impairment and kidney transplantation

Periodic monitoring of serum potassium and creatinine levels is recommended when administering this drug to renal patients. There is no experience regarding the administration of telmisartan in patients with recent kidney transplantation. This medicine does not require dose adjustment in renal patients including those receiving hemodialysis, and is not removed by hemodialysis.

2) Intravascular hypovolemia

Symptomatic hypotension, especially after the first dose, may occur in patients who are volume and/or sodium depleted by vigorous diuretic therapy, dietary salt restriction, diarrhoea or vomiting. Such conditions should be corrected before the administration of this medicine. If hypotension occurs, lay the patient flat and inject intravenous injection of saline solution if necessary.

3) Other conditions with stimulation of the renin-angiotensin-aldosterone system

In patients whose vascular tone and renal function depend predominantly on the activity of the

renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with medicinal products that affect this system has been associated with acute hypotension, hyperazotaemia, oliguria, or rarely acute renal failure. Similar results are expected in patients taking this medicine.

4) Electrolyte imbalance / hyperkalemia

The use of medicinal products that affect the renin-angiotensin-aldosterone system may cause hyperkalaemia. In the elderly, in patients with renal insufficiency, in diabetic patients, in patients concomitantly treated with other medicinal products that may increase potassium levels, and/or in patients with intercurrent events, hyperkalaemia may be fatal. Before considering the concomitant use of medicinal products that affect the renin-angiotensin-aldosterone system, the benefit risk ratio should be evaluated.

The main risk factors for hyperkalaemia to be considered are:

- Diabetes mellitus, renal impairment, age (>70 years)
- Combination with one or more other medicinal products that affect the renin-angiotensin-aldosterone system and/or potassium supplements. Medicinal products or therapeutic classes of medicinal products that may provoke hyperkalaemia are salt substitutes containing potassium, potassium-sparing diuretics, ACE inhibitors, angiotensin II receptor antagonists, non-steroidal anti-inflammatory medicinal products (NSAIDs, including selective COX-2 inhibitors), heparin, immunosuppressives (cyclosporin or tacrolimus), and trimethoprim.
- Intercurrent events, in particular dehydration, acute cardiac decompensation, metabolic acidosis, worsening of renal function, sudden worsening of the renal condition (e.g. infectious diseases), cellular lysis (e.g. acute limb ischemia, rhabdomyolysis, extend trauma).

5) This product should be stopped for 24 hours prior to surgery.

6) In patients with gastrointestinal disorders such as active stomach or duodenal ulcer, gastrometastatic adverse events were more frequent when telmisartan was taken than placebo. Gastrointestinal bleeding was rarely reported in clinical trials and appeared initially in patients with gastrointestinal disease. Therefore, this medicine should be prescribed with caution in patients with gastrointestinal disorders.

7) There is an increased risk of severe hypotension and renal insufficiency when patients with bilateral renal artery stenosis or stenosis of the artery to a single functioning kidney are treated with medicinal products that affect the renin-angiotensin-aldosterone system. When the medicine is administered to patients with either or both renal artery stenosis, the elevation of serum creatinine or serum uric acid nitrogen is predicted similar to ACE inhibitors. In addition, it is recommended to avoid use except for the case where the treatment is inevitable, because telmisartan administration may deteriorate renal function by a reduction in the kidney blood flow and glomerular filtration pressure.

8) Patients with primary aldosteronism generally will not respond to antihypertensive medicinal products acting through inhibition of the renin-angiotensin system. Therefore, the use of telmisartan is not recommended.

9) In patients with diabetes with additional risk for cardiovascular disease (eg, patients with diabetes and coronary artery disease), hypotensive agents such as angiotensin II receptor antagonists or ACE blockers may cause fatal myocardial infarction and unexpected cardiovascular disease, so that the risk of death may increase. Coronary artery disease may not be diagnosed because diabetic patients do not have symptoms of the disease, so proper diagnosis assessment (eg, exercise stress test) for coronary artery disease should be conducted priority in diabetic patients before administration.

10) Geriatric Patients: No initial dosage adjustment is necessary for elderly patients. The pharmacokinetics of telmisartan do not differ between the elderly and those younger than 65 years

11) Pediatric Use: Safety and effectiveness in pediatric patients have not been established. Telmisartan pharmacokinetics have not been investigated in patients <18 years of age.

Rosuvastatin

1) Interstitial lung disease: Exceptional cases of interstitial lung disease have been reported with some statins, especially with long term therapy. Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, statin therapy should be discontinued.

2) Post-marketing cases of fatal and nonfatal hepatic insufficiency were rarely reported in patients taking statin products including rosuvastatin. Treatment should be discontinued immediately if symptoms of serious liver injury and/or hyperbilirubinemia or jaundice occurs during the administration of this medicine. If no other pathogens are found, this medicine should not be re-administered.

3) Increased HbA1c and fasting serum glucose levels have been reported with HMG-CoA reductase inhibitors including rosuvastatin.

4) Diabetes Mellitus: Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9 mmol/L, BMI>30kg/m², raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines. In the JUPITER study, the reported overall frequency of diabetes mellitus was 2.8% in rosuvastatin and 2.3% in placebo, mostly in patients with fasting glucose 5.6 to 6.9 mmol/L.

5) Genetic polymorphisms: Individual polymorphisms of SLCO1B1 c.521CC and ABCG2 c.421AA are associated with a higher rosuvastatin exposure (AUC) compared to the SLCO1B1 c.521TT or ABCG2

c.421CC genotypes. This specific genotyping is not established in clinical practice, but for patients who are known to have these types of polymorphisms, a lower daily dose of rosuvastatin is recommended.

6) Severe cutaneous adverse reactions: Severe cutaneous adverse reactions including Stevens-Johnson syndrome (SJS) and drug reaction with eosinophilia and systemic symptoms (DRESS), which could be life-threatening or fatal, have been reported with rosuvastatin. At the time of prescription, patients should be advised of the signs and symptoms of severe skin reactions and be closely monitored. If signs and symptoms suggestive of this reaction appears, rosuvastatin should be discontinued immediately and an alternative treatment should be considered. If the patient has developed a serious reaction such as SJS or DRESS with the use of rosuvastatin, treatment with this drug must not be restarted in this patient at any time.

7) Genetic polymorphism: Pharmacogenomics: Disposition of rosuvastatin, involves OATP1B1 and other transporter proteins. Higher plasma concentrations of rosuvastatin have been reported in very small groups of patients (n=3 to 5) who have two reduced function alleles of the gene that encodes OATP1B1 (SLCO1B1 521T > C). The frequency of this genotype (i.e., SLCO1B1 521 C/C) is generally lower than 5% in most racial/ethnic groups. The impact of this polymorphism on efficacy and/or safety has not been clearly established.

4. Pediatric Use

Administration of this drug is not recommended because the safety and efficacy of this drug has not been established in children.

5. Use in the Elderly

Excessive hypotension is generally not desirable for the elderly (cerebral infarction may occur). When administered to the elderly, this medicine should be administered with caution while observing the patient's condition.

[Contraindications]

Do not use this drug to the following patients.

- 1) Patients with hypersensitivity to this product or ingredients.
- 2) During pregnancy and lactation and in women of childbearing potential not using appropriate contraceptive measures.

Telmisartan should be used during pregnancy only with caution that this drug can increase risk of spontaneous abortion and cause serious (possibly fatal) harm to an unborn baby.

- 3) Patients with cholestasis, biliary obstructive disorders or severe hepatic impairment. (Telmisartan is mostly eliminated with the bile. These patients can be expected to have reduced hepatic clearance for telmisartan.)

- 4) Patients with active liver disease including unexplained, persistent elevations of serum transaminases and any serum transaminase elevation exceeding 3 times the upper limit of normal (ULN).
- 5) Patients with myopathy.
- 6) Patients with hereditary angioedema or with a history of angioedema from ACE inhibitor or angiotensin II receptor antagonists.
- 7) Patients with severe renal impairment (creatinine clearance <30 ml/min).
- 8) Patients who take telmisartan and cyclosporine together.
- 9) The concomitant use of telmisartan and aliskiren-containing product in patients with diabetes mellitus or renal impairment (GFR<60mL/min/1.73m²)
- 10) The 40mg dose of rosuvastatin is contraindicated in patients with pre-disposing factors for myopathy/rhabdomyolysis. Such factors included:
 - moderate renal impairment (creatinine clearance < 60 mL/min)
 - hypothyroidism
 - personal or family history of hereditary muscular disorders
 - previous history of muscular toxicity with another HMG-CoA reductase inhibitor or fibrate
 - alcohol abuse
 - situations where an increase in plasma levels may occur
 - Asian patients
 - concomitant use of fibrates.
- 11) This product contains lactose, so patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

[Adverse Reaction]

Telmisartan/rosuvastatin combination drug

- 1) According to clinical study of the combination therapy of rosuvastatin and telmisartan administered in 208 patients with hypertension and hyperlipidemia, the tolerability was good and no specific adverse reaction that occurred only in this group was reported, when compared to administration of single-dose telmisartan or rosuvastatin. In the group (82 patients) of combination therapy of rosuvastatin and telmisartan, adverse events related to this drug were as following; ALT increased, AST increased, CPK increased, dizziness and pruritus was reported each in 1 patient, and all adverse drug reactions were mild or moderate. All adverse events that occurred after administration were recorded as below, even if they were not related to the test drug.
 - Digestive system: Dyspepsia, Paraesthesia oral
 - Musculoskeletal and connective tissue disorders: Musculoskeletal stiffness
 - Skin and subcutaneous tissue disorders: Acne, Pruritus
 - Ear and labyrinth disorders: Vertigo
 - Infections and infestations: Nasopharyngitis
 - Investigations: Aspartate aminotransferase increased, Alanine aminotransferase increased, Blood

creatine phosphokinase increased, Blood potassium increased, Glucose urine present

2) Postmarket survey results in Korea

2-1) In Korea, 658 patients were surveyed for 6 years, and the incidence rate of adverse events was reported to be 21.12% (139/658 patients, 187 cases). The serious adverse drug reaction and unexpected adverse drug reactions were not reported.

2-2) In Korea, 1,503 patients were surveyed for 6 years, and the incidence rate of adverse events was reported to be 11.04% (166/1,503 patients, 292 cases). The serious adverse drug reaction was not reported, and unexpected adverse drug reactions were listed in the table below according to the frequency.

Frequency for reactions	System organ class	Unexpected adverse drug reactions 1.00% (15/1,503 patients, 18 cases)
Uncommon (0.1% to 1%)	Urinary system disorders	Painful urination
Rare (0.01% to 0.1%)	Respiratory system disorders	Rhinitis, Oropharyngeal discomfort, Pulmonary disease
	Gastrointestinal disorders	Gastrointestinal disease of unspecified, Benign neoplasms of gastrointestinal tract, Hemorrhoids
	Musculoskeletal disorders	Skeletal pain, Osteoporosis
	Central & peripheral nervous system disorders	Numbness, Dyskinesia
	Vascular disorders	Atherosclerosis
	Urinary system disorders	Frequent urination
	Hearing and vestibular disorders	Tinnitus, Hearing loss
	Body as a whole-general disorders	Halitosis
	Liver and biliary tract disease	Fatty liver

The safety profile below is based on data from clinical trials and post-marketing experiences of the individual components, telmisartan and rosuvastatin.

Information linked to telmisartan

1) The overall incidence of adverse reactions reported with telmisartan was usually comparable to placebo (41.4% vs 43.9%) in placebo controlled trials in patients treated for hypertension. The incidence of adverse reactions was not dose related and showed no correlation with gender, age or race of the patients. The safety profile of telmisartan in patients treated for the reduction of cardiovascular morbidity was consistent with that obtained in hypertensive patients.

The adverse reactions listed below have been accumulated from controlled clinical trials in patients treated for hypertension and from post-marketing reports. The listing also takes into account serious adverse reactions and adverse reactions leading to discontinuation reported in three clinical long-

term studies including 21,642 patients treated with telmisartan for the reduction of cardiovascular morbidity for up to six years.

Adverse reactions have been ranked under headings of frequency using the following convention: uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$).

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

System organ class	Frequency for reactions	
	Uncommon	Rare
Infection and Infestations	Upper respiratory tract infection (including pharyngitis, sinusitis), Urinary tract infection (including cystitis)	Sepsis (including a serious result)
Blood and Lymphatic System Disorders	Anemia	Thrombocytopenia, Eosinophilia
Immune System Disorders		Hypersensitivity, Anaphylactic reaction
Metabolism (Metabolic system) and Nutrition Disorders	Hyperkalemia	Hypoglycemia
Psychiatric Disorders	Depression, Insomnia	Agitation
Nervous System Disorders	Syncope	
Eye Disorders		Heteroptics
Ear and labyrinth disorders	Vertigo	
Cardiac Disorders	Bradycardia	Tachycardia
Vascular Disorder	Hypotension*, Orthostatic Hypotension	
Respiration, Thoracic, and Mediastinum	Dyspnea	
Gastro-intestinal tract	Abdominal pain, Diarrhea, Dyspepsia, Abdominal distension, Vomiting	Abdominal discomfort, Xerostomia
Hepato-Biliary Disorders		Hepatic function abnormal /Hepatopathy**
Skin and subcutaneous tissue disorders	Pruritus, , hyperhidrosis, rash	Angioedema (also with fatal outcome), eczema, erythema, urticaria, drug eruption, toxic skin eruption
Musculoskeletal and connective tissue disorders	Back pain (e.g. sciatica), muscle spasms, myalgia	Arthralgia, pain in extremity, tendon pain (tendinitis like symptoms)
Renal and urinary disorders	Renal impairment including acute renal failure	
General disorders and administration site	Chest pain, asthenia (weakness)	Influenza-like illness

conditions		
Investigations	Blood creatine phosphokinase increased	Haemoglobin decreased, blood uric acid increased, hepatic enzyme increased, blood creatine phosphokinase increased

* This adverse reaction was reported as common in patients with controlled blood pressure who were treated with telmisartan for the reduction of cardiovascular morbidity on top of standard care.

** Most cases of hepatic function abnormal / liver disorder from post-marketing experience occurred in patients in Japan, who are more likely to experience these adverse reactions.

2) Metabolism disorders: cases of hypoglycemia have been rarely reported. (in diabetic patients) Patients should be monitored carefully. If the helplessness, hunger sensation, cold sweat, essential tremor, busy brain, convulsion, and decreased consciousness, it is recommended to stop the treatment and take proper action.

3) Angioedema: As swelling of your face, lips, tongue, or throat, the case of difficulty breathing or swallowing was reported. Patients should be monitored carefully. If the adverse events occurs, it is recommended to stop the treatment immediately and take proper action.

4) Hyperkalemia: Telmisartan may cause hyperkalemia. Use with caution in patients, and stop the treatment immediately and take proper action if the adverse events occurs.

5) Renal dysfunction: the case of renal failure were reported. Use with caution in patients, and stop the treatment immediately and take proper action if the adverse events occurs.

6) Shock, syncope, and loss of consciousness: Since syncope and loss of consciousness from hypotension, and shock may occur with telmisartan, use with caution in patients and take proper action if cold sensation, vomiting, and loss of consciousness occur. Specially, in patients with a hemodialysis, on low-salt diet, or taking diuretics, previously take low doses and titrate slowly with caution if necessary.

7) Dyshepatia, jaundice: Dyshepatia such as AST increased, ALT increased, ALP increased, LDH increased, and etc. and jaundice may occur with telmisartan. Use with caution in patients, and stop the treatment and take proper action if the adverse events occurs.

8) Anaphylaxis symptom: Allergic reactions including dyspnea, hypotension, or laryngeal edema may occur. Use with caution in patients, and stop the treatment immediately and take proper action if the adverse events occurs.

9) Interstitial lung disease (unknown frequency): Symptoms of interstitial lung disease like fever, cough, difficulty breathing, chest x-ray abnormalities may occur. In this case, stop the treatment and take proper action.

10) Rhabdomyolysis (unknown frequency): Rhabdomyolysis that the presenting features are myalgias, weakness, an elevated CK, and serum and urine myoglobin increased may occur. Use with caution in patients and take proper action like steroid therapy.

11) In addition to the above adverse reactions, palpitation, fatigue and lack of effects were reported.

12) As the result of the large-scale post-marketing study on 15,601 hypertension patients in Korea, the overall incidence of adverse reactions was 1.54% (240cases/15,601cases), and the drug-related

adverse reactions was 1.2% (183cases/15,601cases). The most common ADR was headache 0.38% (60cases/15,601cases) and followed dizziness 0.2% (33cases/15,601cases), coughing 0.13% (21cases/15,601cases), indigestion 0.07% (11cases//15,601cases), palpitation 0.06% (10cases/15,601cases). One case reported serious adverse reaction of arrhythmia. The following adverse reactions listed below, with no established causality, had lower incidence (*) than placebo in the pre-clinical trial or were the new adverse events that were not appeared in preclinical trials. (Numbers in parentheses represent the number of cases)

- (1) Central and peripheral nervous system: headache*(60), facial spasm(2), paresthesia(2), paralysis(1)
- (2) Respiratory system: coughing*(21), hyperpnea(1), pneumonia(1)
- (3) Autonomic nerve: telangiectasia(5), erectile dysfunction(1), anorexia(1), arrhythmia(1)
- (4) Whole body: impotent(4), edema(3)
- (5) Psychiatric disorders: sexual desire disorder(2)
- (6) Skin and subcutaneous tissue disorders: facial edema(1), rash(1), alopecia(1)
- (7) Sensory system: conjunctivitis(1), tinnitus(1)
- (8) Platelet hemorrhage and coagulation disorders: petechia(1)

Among these, the incidence of adverse reactions in patients with renal disease was 0.9% (4cases/445cases), and headache, dizziness, myalgia, and dermatitis were reported. The incidence of adverse reactions in patients with hepatopathy was 2.0% (6cases/296cases), and dizziness, indigestion, coughing, anorexia, impotent were reported.

Information linked to Rosuvastatin

1) The reported adverse events were generally mild and transient. In controlled clinical trials, less than 4% of rosuvastatin-treated patients were withdrawn due to adverse reactions.

The frequencies of adverse reactions are ranked according to the following convention: Common ($\geq 1/100$ to $< 1/10$); Uncommon ($\geq 1/1,000$ to $< 1/100$); Rare ($\geq 1/10,000$ to $< 1/1000$); Very rare ($< 1/10,000$); Not known (cannot be estimated from the available data).

System organ class	Frequency for reactions		
	Common	Uncommon	Rare
Immune system			Hypersensitivity reactions including angioedema
Endocrine system	Diabetes mellitus ¹⁾		
Nervous system	Headache, dizziness		
Gastrointestinal relationship	Constipation, area, abdominal pain		Pancreatitis
Skin and subcutaneous tissue		Pruritus, Rash and urticaria	

Musculoskeletal and connective tissue	Myalgia		Myopathy (including myositis) Rhabdomyolysis
Whole body	Asthenia		

¹⁾ In the JUPITER study, the most frequently reported adverse events (fasting blood glucose ≥ 5.6 to 6.9 mmol/L, 2.8% in rosuvastatin, 2.3% in placebo)

As with other HMG-CoA reductase inhibitors, the incidence of adverse drug reactions tends to be dose dependent.

2) Renal effects: Proteinuria, detected by dipstick testing and mostly tubular in origin, has been observed in patients treated with rosuvastatin. Shifts in urine protein from none or trace to ++ or more were seen in <1% of patients at some time during treatment with 10 and 20 mg, and in approximately 3% of patients treated with 40 mg. A minor increase in shift from none or trace to + was observed with the 20 mg dose. In most cases, proteinuria decreases or disappears spontaneously on continued therapy. Review of data from clinical trials and post-marketing experience to date has not identified a causal association between proteinuria and acute or progressive renal disease.

3) Skeletal muscle effects: Effects on skeletal muscle e.g. myalgia, myopathy (including myositis) and, rarely, rhabdomyolysis with and without acute renal failure have been reported in rosuvastatin-treated patients with all doses and in particular with doses > 20 mg.

A dose-related increase in CK levels has been observed in patients taking rosuvastatin; the majority of cases were mild, asymptomatic and transient. If CK levels are elevated ($>5 \times \text{ULN}$), treatment should be discontinued.

4) Liver effects: As with other HMG-CoA reductase inhibitors, a dose-related increase in transaminases has been observed in a small number of patients taking rosuvastatin; the majority of cases were mild, asymptomatic and transient.

5) Experience after overseas marketing

In addition to the above adverse events, the following adverse events were reported during postmarketing investigations.

- Nervous system: Very rarely multiple neuropathy, amnesia
- Respiratory and thoracic: cough, dyspnea (frequency unknown)
- Gastrointestinal: diarrhea (frequency unknown)
- Hematologic disorder: thrombocytopenia (frequency unknown)
- Hepatobiliary system: very rare jaundice, hepatitis, rarely increased transaminase
- Skin and subcutaneous tissue disorders: Stevens-Johnson syndrome (frequency unknown)
- Musculoskeletal: Very rarely arthritis, immune-mediated necrotizing myopathy (frequency unknown)
- Renal: Very rarely hematuria

- Other: edema (frequency unknown)

The following adverse reactions have been reported with some statins.

- Psychiatric: Depression, sleep disorders (including sleeping and nightmares) (frequency unknown)
- Respiratory system: Exceptional cases such as interstitial lung disease especially during long term administration.
- Reproductive system: Sexual dysfunction, gynecomastia (frequency unknown)
- Hepatobiliary system: Fatal and nonfatal liver failure

Post-marketing cognitive impairment with statin use is rarely reported. (eg. memory loss, forgetfulness, amnesia, memory impairment, confusion). These cognitive disorders have been reported in all statin drugs. These reports are generally inaccurate and reversible after discontinuation of medication, and there are deviations in symptom onset (1 day to several years) and symptom improvement (median 3 weeks).

6) Postmarket survey results in Korea

In Korea, 3,081 patients were surveyed for 6 years, and the incidence rate of harmful cases was 10.06% (310 cases, 415 cases). The most common ADR was headache 0.78% (24 persons, 24 cases) and followed dizziness 0.75% (23 patients, 23 cases), ALT increased 0.58% (18 patients, 18 cases), chest pain, cough and myalgia were reported in 0.49% (15 patients, 15 cases). And the drug-related adverse reactions was 2.92% (90 patients, 106 cases). The most common reported ADR was ALT increased 0.55% (17 patients, 17 cases) and followed myalgia 0.42% (13 patients, 13 cases), headache 0.39% (12 patients, 12 cases), CK increased 0.29% (9 patients, 9 cases), dizziness 0.26% (8 patients, 8 cases), constipation and AST increased 0.16% (5 patients, 5 cases) each, asthenia and joint pain 0.13% (4 patients, 4 cases) each, fatigue and anesthesia 0.10% (3 patients, 3 cases), sensory abnormality, chest discomfort, nausea, abdominal pain, diarrhea, anorexia, abdominal distension, itching and liver function test disorder 0.06% (2 patients, 2 cases) each, syncope, systemic pain, muscle spasms, gout, and erectile dysfunction 0.03% (1 patient, 1 case) each. One case of joint pain and myalgia was reported as a serious and unexpected drug adverse reaction. The unexpected adverse drug reactions that did not appear before the market were joint pain 0.13% (4 patients, 4 cases), fatigue and numbness were 0.10% (3 patients, 3 cases) each, sensory abnormality, chest discomfort, anorexia, abdominal distension and liver function test disorder 0.06% (2 patients, 2 cases) each, syncope, systemic pain, muscle spasm, gout, and erectile dysfunction were 0.03% (1 patients, 1 case) each. There were 98 adverse events reported voluntarily during the review period, of which two acute renal failure cases reported and single case of each oliguria, thrombocytopenia, and elevated serum creatinine were reported as serious and unexpected adverse drug reactions.

[Interaction with other medicinal products and other forms of interaction]

Drug interaction studies with telmisartan 80 mg and rosuvastatin 20 mg in healthy volunteers showed that the C max of telmisartan in the presence of rosuvastatin increased about 1.35 fold and the AUC increased about 1.17 fold. The Cmax of rosuvastatin in the presence of telmisartan increased

about twice and the AUC increased about 1.18 times. Studies of drug interactions with other medications and telmisartan/rosuvastatin combination agents have never been conducted, but individual studies on telmisartan and rosuvastatin have been performed as follows.

Telmisartan

- 1) The double blockade of the renin-angiotensin-aldosterone system (RAAS) by the combined administration of angiotensin II receptor antagonist (ARB), ACE inhibitor or aliskiren compared to single therapy of these drugs was related to increasing risk of hypotension, syncope, hyperkalemia and renal function changes(including acute renal failure). In patients who taken concomitant use of telmisartan and other medicines acting on RAAS, blood pressure, renal function and electrolyte should be closely monitored. Do not co-administer aliskiren with telmisartan in patients with diabetes or moderate to severe renal impairment (GFR <60 mL/min/1.73 m²).
- 2) It may increase the blood pressure lowering effect of other hypotensive medicines. No other clinically relevant interactions have been identified.
- 3) Coadministration of telmisartan did not result in a clinically significant interactions with warfarin, hydrochlorothiazide, glibenclamide, ibuprofen, acetaminophen, simvastatin or amlodipine.
- 4) Digoxin: When telmisartan was co-administered with digoxin, median increases in digoxin peak plasma concentration (49%) and in trough concentration (20%) were observed. When initiating, adjusting, and discontinuing telmisartan, monitor digoxin levels in order to maintain levels within the therapeutic range.
- 5) Ramipril: In one study the co-administration of telmisartan and ramipril led to an increase of up to 2.5 fold in the AUC₀₋₂₄ and C_{max} of ramipril and ramiprilat. When combined with telmisartan and ramipril, the effect can be enhanced due to the pharmacodynamic effect and the increased exposure of the ramipril / ramiprilat. Concomitant use of ramipril and telmisartan is not recommended.
- 6) Lithium: Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with angiotensin converting enzyme inhibitors, and with angiotensin II receptor antagonists, including telmisartan. If use of the combination proves necessary, careful monitoring of serum lithium levels is recommended.
- 7) Telmisartan is not metabolized by the CYP-450 system and had no effects in vitro on CYP-450 enzymes, except for some inhibition of CYP2C19. Telmisartan is not expected to interact with drugs that inhibit CYP-450 enzymes; it is also not expected to interact with drugs metabolized by CYP-450 enzymes, except for possible inhibition of the metabolism of drugs metabolized by CYP2C19.
- 8) Orthostatic hypotension may be aggravated by baclofen, amifostine, alcohol, barbiturates, drugs, and antidepressants.
- 9) Nonsteroidal anti-inflammatory drugs (NSAIDs): NSAIDs (i.e. acetylsalicylic acid (> 3 g / day) at anti-inflammatory dosage regimens, COX-2 inhibitors and non-selective NSAIDs) may reduce the

antihypertensive effect of angiotensin II receptor antagonists. In some patients with compromised renal function (e.g. dehydrated patients or elderly patients with compromised renal function), the co-administration of angiotensin II receptor antagonists and agents that inhibit cyclo-oxygenase may result in further deterioration of renal function, including possible acute renal failure, which is usually reversible. Therefore, the combination should be administered with caution, especially in the elderly. Patients should be adequately hydrated and consideration should be given to monitoring of renal function after initiation of concomitant therapy and periodically thereafter.

10) Potassium sparing diuretics or potassium supplements: Angiotensin II receptor antagonists such as telmisartan, attenuate diuretic induced potassium loss. Potassium sparing diuretics e.g. spironolactone, eplerenone, triamterene, or amiloride, potassium supplements, or potassium-containing salt substitutes may lead to a significant increase in serum potassium. If concomitant use is indicated because of documented hypokalaemia they should be used with caution and with frequent monitoring of serum potassium.

11) Corticosteroids (systemic route): antihypertensive effect may be reduced when co-administered with this drug.

12) Diuretics (thiazides or loop diuretics): Prior treatment with high dose diuretics such as furosemide (loop diuretic) and hydrochlorothiazide (thiazide diuretic) may result in volume depletion and in a risk of hypotension when initiating therapy with telmisartan.

Rosuvastatin

1) Effect of co-administered medicinal products on rosuvastatin

Results from *in vitro* and *in vivo* studies show that rosuvastatin is neither an inhibitor nor an inducer of cytochrome P450 isoenzymes. Rosuvastatin is a substrate for certain transporter proteins including the hepatic uptake transporter OATP1B1 and efflux transporter BCRP. Concomitant administration of rosuvastatin with medicinal products that are inhibitors of these transporter proteins may result in increased rosuvastatin plasma concentrations and an increased risk of myopathy. (See table below).

<Effect of co-administered medicinal products on rosuvastatin exposure (AUC) from published clinical results>

Interacting drug dose regimen	Rosuvastatin dose regimen	Changes in rosuvastatin AUC
Ciclosporin 75~200 mg twice daily, 6 months	10 mg once daily, 10 days	7.1-fold ↑
Darolutamide 600 mg twice daily, 5 days	5 mg, single dose	5.2-fold ↑
Regorafenib 160mg once daily, 14 days	5 mg, single dose	3.8-fold ↑
Atazanavir 300 mg/ritonavir 100 mg once	10 mg, single dose	3.1-fold ↑

daily, 8 days		
Velpatasvir 100mg once daily	10 mg, single dose	2.7-fold ↑
Ombitasvir 25mg/paritaprevir 150mg/ ritonavir 100mg once daily/dsabuvir 400mg twice daily, 14 days	5 mg, single dose	2.6-fold ↑
Elbasvir 50mg/grazoprevir 200mg once daily, 11 days	10 mg, single dose	2.3-fold ↑
Glecaprevir 400mg/pibrentasvir 120mg once daily, 7 days	5 mg once daily, 7 days	2.2-fold ↑
Lopinavir 400 mg/ritonavir 100 mg twice daily, 17 days	20 mg once daily, 7 days	2.1-fold ↑
Clopidogrel 300 mg loading, followed by 75 mg at 24 hours	20 mg, single dose	2-fold ↑
Gemfibrozil 600 mg twice daily for 7 days	80 mg, single dose	1.9-fold ↑
Eltrombopag 75 mg single dose, 10 days	10 mg, single dose	1.6-fold ↑
Darunavir 600 mg/ritonavir 100 mg twice daily, 7 days	10 mg once daily, 7 days	1.5-fold ↑
Tipranavir 500 mg/ritonavir 200 mg twice daily, 11 days	10 mg, single dose	1.4-fold ↑
Dronedarone 400 mg twice daily	10 mg	1.4-fold ↑
Itraconazole 200 mg once daily, 5 days	10 mg, single dose 80 mg, single dose	1.4-fold ↑ 1.3-fold ↑
Ezetimibe 10 mg once a day for 14 days	10 mg once daily , 14 days	1.2-fold ↑
Fosamprenavir 700 mg/ritonavir 100 mg Twice daily, 8 days	10 mg, single dose	No change
Aleglitazar 0.3 mg, 7 days	40 mg, 7 days	No change
Silymarin 140 mg 3 times daily, 5 days	10 mg, single dose	No change
Fenofibrate 67 mg 3 times daily, 7 days	10 mg, 7 days	No change
Rifampin 450 mg once daily, 7 days	20 mg, single dose	No change
Ketoconazole 200 mg twice daily, 7 days	80 mg, single dose	No change
Fluconazole 200 mg once daily, 11 days	80 mg, single dose	No change
Erythromycin 500 mg 4 times daily, 7 days	80 mg, single dose	20% ↓
Baicalin 50 mg 3 times daily, 14 days	20 mg, single dose	47% ↓

Other medicinal products

- Antacids: The simultaneous dosing of rosuvastatin with an antacid suspension containing

aluminium and magnesium hydroxide resulted in a decrease in rosuvastatin plasma concentration of approximately 50%. This effect was mitigated when the antacid was dosed 2 hours after rosuvastatin. The clinical relevance of this interaction has not been studied.

2) Effect of rosuvastatin on co-administered medicinal products

- Warfarin: Warfarin is not significantly affected by pharmacokinetics when used in combination with rosuvastatin. However, as with other HMG-CoA reductase inhibitors, the initiation of treatment or dosage up-titration of rosuvastatin in patients treated concomitantly with vitamin K antagonists (e.g. warfarin or another coumarin anticoagulant) may result in an increase in International Normalised Ratio (INR). Discontinuation or down-titration of rosuvastatin may result in a decrease in INR. In such situations, appropriate monitoring of INR is desirable.
- Ciclosporin: Concomitant administration with rosuvastatin and ciclosporin did not affect plasma concentrations of ciclosporin.
- Fenofibrate / Other fibrates: Based on data from specific interaction studies no pharmacokinetic relevant interaction with fenofibrate is expected, however a pharmacodynamic interaction may occur. Gemfibrozil, fenofibrate, other fibrates and lipid lowering doses (> or equal to 1g/day) of niacin (nicotinic acid) increase the risk of myopathy when given concomitantly with HMG-CoA reductase inhibitors, probably because they can produce myopathy when given alone. The 40 mg dose is contraindicated with concomitant use of a fibrate. These patients should also start with the 5 mg dose.
- Oral contraceptives: Concomitant use of rosuvastatin and an oral contraceptive resulted in an increase in ethinyl estradiol and norgestrel AUC of 26% and 34%, respectively. These increased plasma levels should be considered when selecting oral contraceptive doses. There are no pharmacokinetic data available in subjects taking concomitant rosuvastatin and HRT and therefore a similar effect cannot be excluded. However, the combination has been extensively used in women in clinical trials and was well tolerated.
- Effects on other drugs: Based on data from specific interaction studies no clinically relevant interaction with digoxin or ezetimibe is expected.

[Fertility, pregnancy and lactation]

Telmisartan

- 1) Fertility: The effect of telmisartan on human fertility has not been studied. In preclinical studies with female and male rats, no effects of Micardis on male and female fertility were observed.
- 2) The use of angiotensin II receptor antagonists is not recommended during the first trimester of pregnancy. Because of the mechanism of action of angiotensin II receptor antagonists, the risk to the fetus cannot be ignored. Exposure to angiotensin II receptor antagonist therapy during the second and third trimesters is known to be associated with hypothermia of the fetus and newborn,

hyperkalemia, neonatal skull underdevelopment, urinary loss and/or anuria or fetal and neonatal damage including reversible or irreversible renal failure, intrauterine growth retardation and death. Oligohydramnios which is suspected to be due to decreased fetal renal function was reported and was related to fetal limb development, craniofacial anomaly, and pulmonary growth failure. Based on retrospective data, the use of ACE inhibitors in the first trimester was associated with a potential risk of birth defects. In addition, it is not clear whether it is due to telmisartan, but it is reported that immature, intrauterine growth retardation, and patent ductus arteriosus were observed. Telmisartan like other drugs directly acting on the renin-angiotensin-aldosterone system should not be used in pregnant women or in women planning to become pregnant. When pregnancy is diagnosed, treatment with telmisartan should be stopped immediately. If a medicine that affects the renin-angiotensin-aldosterone system is prescribed in women who are likely to become pregnant, the doctor should tell patients about the potential risks of this medicine during pregnancy. If pregnant women are exposed to angiotensin II receptor antagonists after the second trimester, fetal renal function and cranial ultrasonography are recommended. Also, adequate urination, hyperkalemia, and blood pressure should be investigated in newborns who have experienced exposure to this medicine in the uterus.

3) Because no information is available regarding the use of telmisartan during breast-feeding and telmisartan has been shown to pass through the breast milk of rats, telmisartan is not recommended and alternative treatments with better established safety profiles during breast-feeding are preferable, especially while nursing a newborn or preterm infant. Discontinue breastfeeding, if telmisartan is administered.

Rosuvastatin

Rosuvastatin is contraindicated in pregnancy and lactation. Women of child bearing potential should use appropriate contraceptive measures. Since cholesterol and other products of cholesterol biosynthesis are essential for the development of the foetus, the potential risk from inhibition of HMG-CoA reductase outweighs the advantage of treatment during pregnancy. Animal studies provide limited evidence of reproductive toxicity. If a patient becomes pregnant during use of this product, treatment should be discontinued immediately. Rosuvastatin is excreted in the milk of rats. There are no data with respect to excretion in milk in humans.

[Overdose]

Telmisartan

There is limited information available for telmisartan with regard to overdose in humans. The most prominent manifestations of telmisartan overdose were hypotension and tachycardia; bradycardia (due to parasympathetic nerve stimulation), dizziness, and acute renal failure have also been

reported. If treatment hypotensive symptoms occur, supplementary treatment should be performed. Telmisartan is not removed by haemodialysis. The patient should be closely monitored, and the treatment should be symptomatic and supportive. Management depends on the time since ingestion and the severity of the symptoms. Suggested measures include induction of emesis and/or gastric lavage. Activated charcoal may be useful in the treatment of overdose. Serum electrolytes and creatinine should be monitored frequently. If hypotension occurs, the patient should be placed in a supine position, with salt and volume replacements given quickly.

Rosuvastatin

There is no specific treatment in the event of overdose. In the event of overdose, the patient should be treated symptomatically and supportive measures instituted as required. Liver function and CK levels should be monitored. Haemodialysis is unlikely to be of benefit.

[Effects on ability to drive and use machines]

Studies to determine the effect of this medicine on the ability to drive and use machines have not been conducted. However, when driving vehicles or operating machinery it should be taken into account that dizziness or drowsiness may occasionally occur when undergoing antihypertensive therapy such as telmisartan.

[Preclinical studies]

Telmisartan

In preclinical safety studies, doses producing exposure comparable to that in the clinical therapeutic range caused reduced red cell parameters (erythrocytes, haemoglobin, haematocrit), changes in renal haemodynamics (increased blood urea nitrogen and creatinine), as well as increased serum potassium in normotensive animals. In dogs, renal tubular dilation and atrophy were observed. Gastric mucosal injury (erosion, ulcers or inflammation) also was noted in rats and dogs. These pharmacologically-mediated undesirable effects, known from preclinical studies with both angiotensin converting enzyme inhibitors and angiotensin II receptor antagonists, were prevented by oral saline supplementation. In both species, increased plasma renin activity and hypertrophy/hyperplasia of the renal juxtaglomerular cells were observed. These changes, also a class effect of angiotensin converting enzyme inhibitors and other angiotensin II receptor antagonists, do not appear to have clinical significance. No clear evidence of a teratogenic effect was observed, however at toxic dose levels of telmisartan an effect on the postnatal development of the offsprings such as lower body weight and delayed eye opening was observed. There was no evidence of mutagenicity and relevant clastogenic activity in *in vitro* studies and no evidence of carcinogenicity in rats and mice.

Rosuvastatin

- 1) Preclinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, genotoxicity and carcinogenicity potential. Reproductive toxicity was evident in rats, with reduced litter sizes, litter weight and pup survival observed at maternally toxic doses, where systemic exposures were several times above the therapeutic exposure level.
- 2) Pharmacokinetic studies show an approximate 2-fold elevation in median AUC and C_{max} in Asian subjects (Japanese, Chinese, Filipino, Vietnamese and Koreans) compared with Caucasians. A population pharmacokinetic analysis revealed no clinically relevant differences in pharmacokinetics between Caucasian and Black groups.

[List of excipients]

Sodium hydroxide, Meglumine, Mannitol, Povidone, Microcelac 100, Calcium glycerophosphate, Crospovidone, Ferric oxide red, Colloidal silicon dioxide, Sodium stearyl fumarate, Magnesium stearate, Hypromellose, Polyethylene glycol 400, Talc, Polyethylene glycol 6000, Titanium dioxide, Ethanol, Purified water, Ferric oxide yellow (only in Duowell Tablet 40/10mg and Duowell Tablet 80/10mg)

[Storage Condition]

Store below 30°C.

[Shelf Life]

36 months. See "Exp. Date" indicated on the outer box.

[Packing Unit]

30 tablets

[Product Registration Holder]

Eucogen Pharma Sdn Bhd
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